



# Don't Be A Loser!

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of minimum input voltage, dropout voltage will be 0.5V, and  $V_{IN}$  should be higher than 3.0V when  $V_{OUT} = 2.5V$ . But for  $V_{OUT} = 1.2V$ ,  $V_{IN}$  cannot be lower than 2.5V because minimum input voltage for internal circuit is 2.5V. As a result, dropout

voltage in this case is limited as  $1.3V (= 2.5V - 1.2V)$  even though typical dropout voltage specification is 0.5V.

NMOSFET LDO has a different operating mechanism to overcome the restriction of low voltage conver-

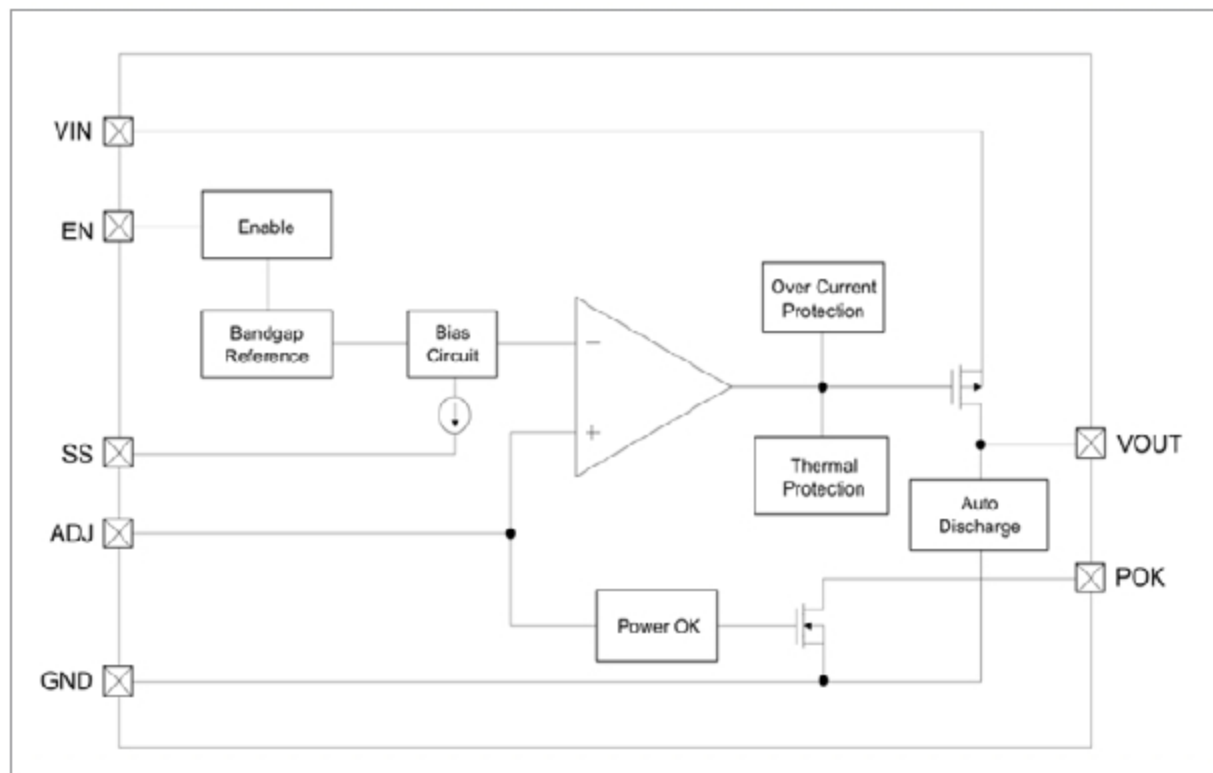


Fig. 5: Simple block diagram of LDO regulator

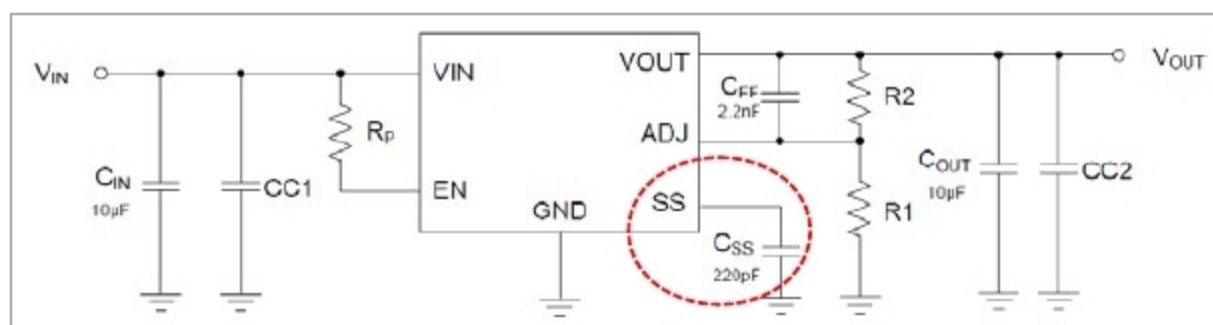


Fig. 6: Application circuit of the external programmable soft start

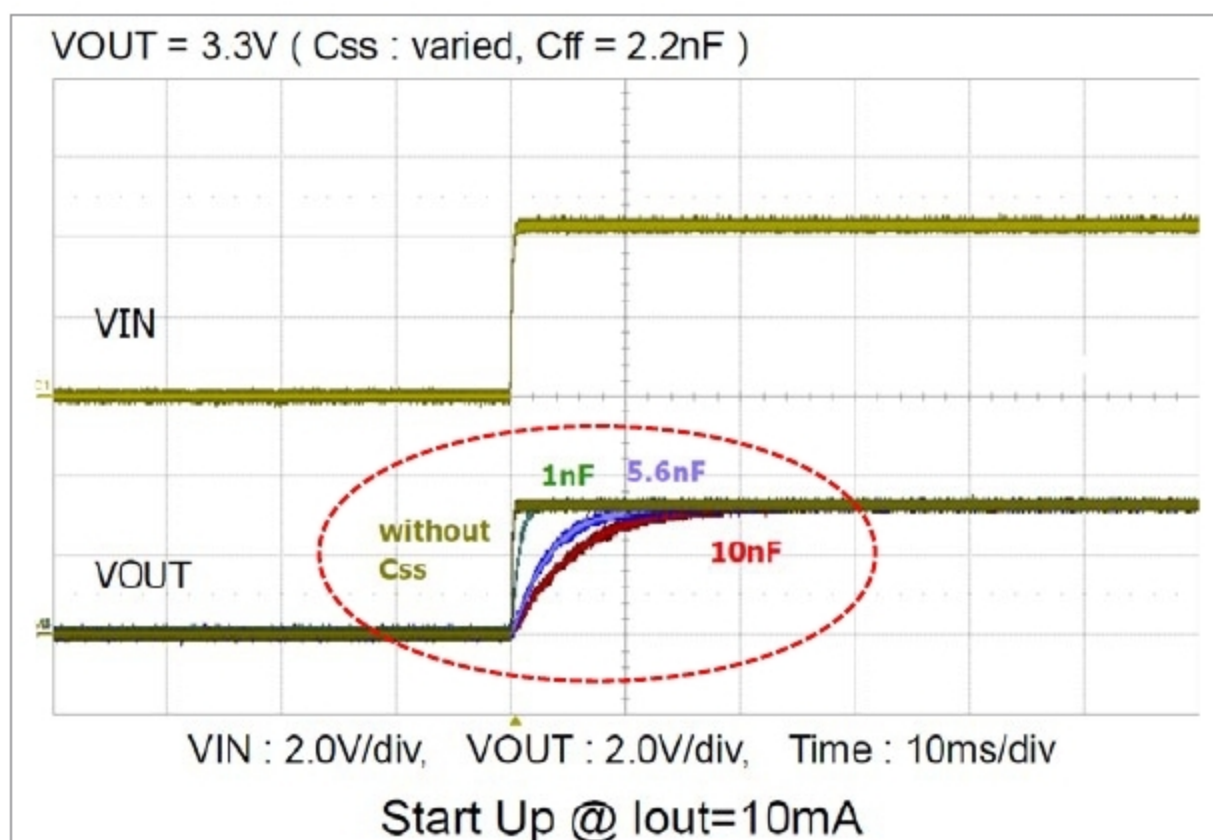


Fig. 7: Rise time of LDO with respect to external  $C_{SS}$